

Autoimmune Diseases and Oral Complications

Integrated Biomedical and Clinical Sciences 2: OMFS 513

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Mission: To advance the science and art of health through education, service, scholarship and social accountability
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Learning Outcomes & Competency Statements

Learning Outcomes

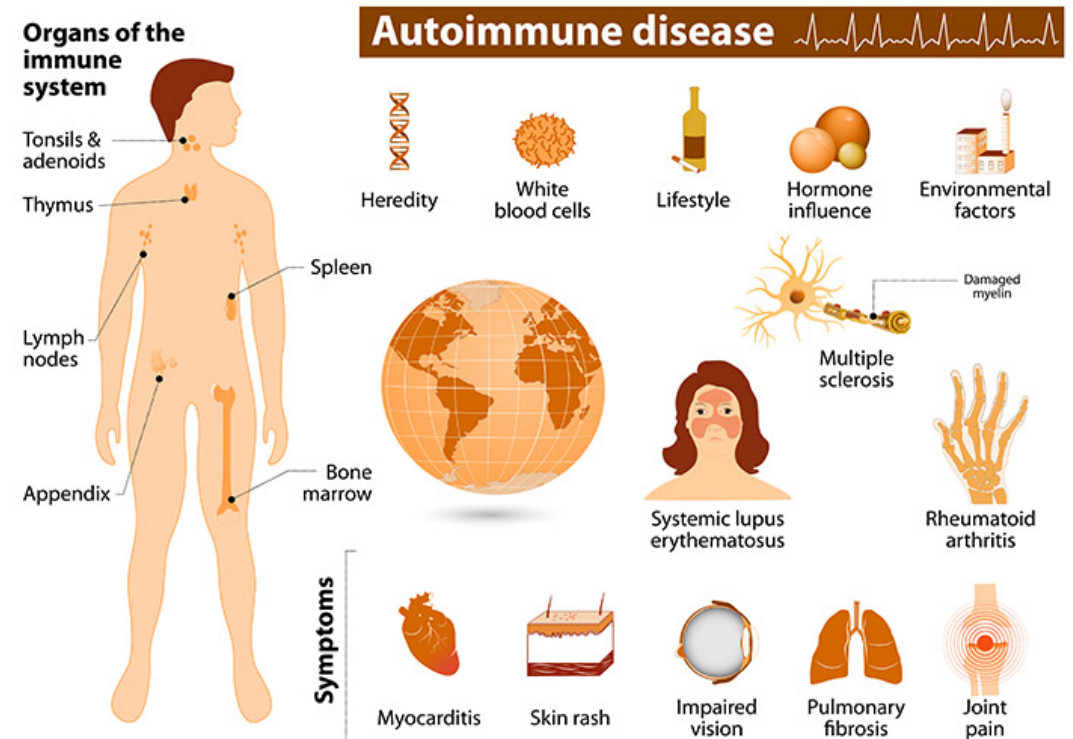
1. Learn about the pathophysiology of selected autoimmune disorders.
2. Review the systemic manifestations of the selected autoimmune diseases.
3. Discuss the impact of autoimmune diseases on the oral cavity and dental treatment.

Competency statements met

1.3, 2.1, 3.1, 4.4, and 5.1

Autoimmune Disease

- Connective tissue disease
- Immune system attacks the host
- Genetic and environmental factors
- Local and widespread disease



Sjogren's Syndrome

- Autoimmune attack on the exocrine glands
- **Clinical**
 - Female predilection
 - **Primary (Sicca Syndrome)**
 - Keratoconjunctivitis
 - Xerostomia
 - Parotid gland enlargement
 - **Secondary**
 - Sicca syndrome in conjunction with another autoimmune disease.
 - Rheumatoid arthritis: most common secondary autoimmune disease



Rischmueller M. et al Best Prac Res Clin Rheum (2016)

Role of the Immune System in Sjogren Syndrome

Sjögren's Disease Explained: The Immune System and What Goes Wrong

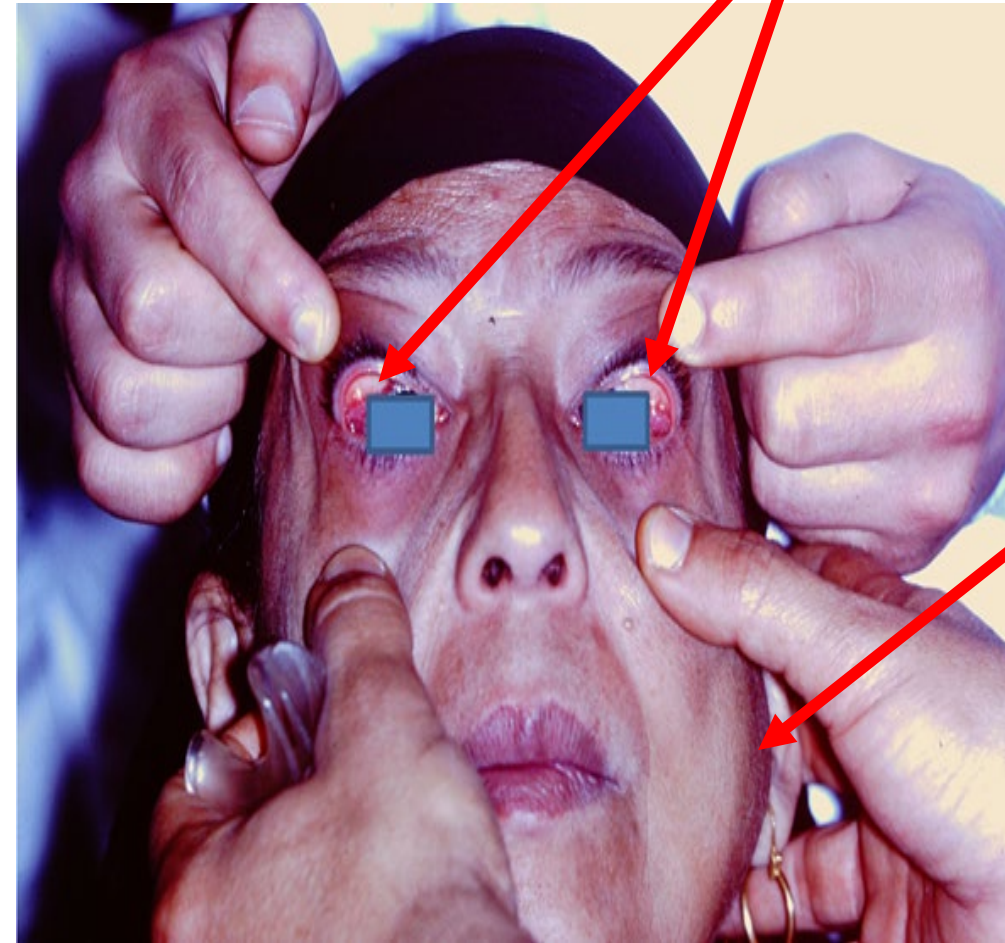
Studies have shown:

1. Hyperactivity of the innate immune system
 - Dendritic Cells, macrophages, and salivary gland epithelial cell release cytokines which initiate the inflammatory process.
 - Enlargement of the affected glands
2. Involvement of the adaptive immune system
 1. Change in the ratio of CD4 helper cells
 2. Increase infiltrate of T lymphocytes in the affected glandular tissue
 3. B-cell dysfunction and secretion of self-reactive antibodies

Sjogren's Syndrome



Bilateral enlargement of the parotid glands



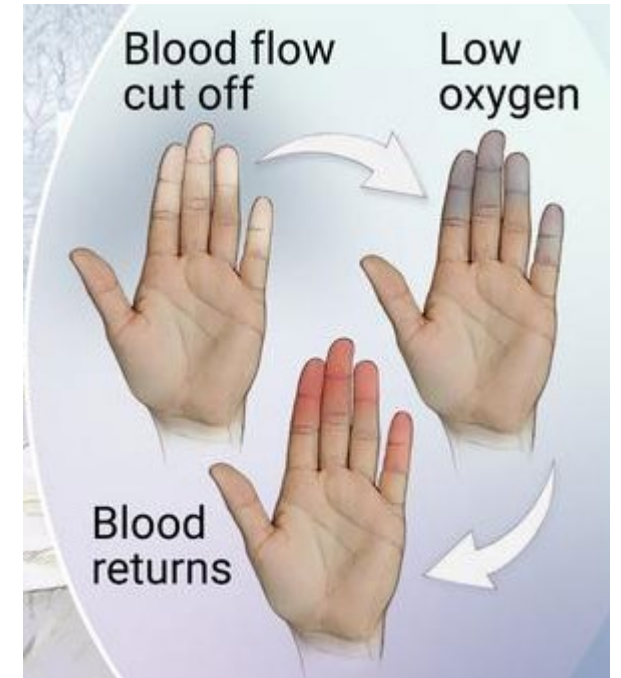
Bilateral enlargement of the lacrimal glands

Enlargement of the parotid gland

Courtesy of Dr. Edmund Cataldo

Sjogren's Syndrome

- Dryness of the nasal mucosa
- Laryngeal changes
- Purpura of the legs
- **Raynaud's phenomenon**
- Autoimmune thyroiditis (Hashimoto's Thyroiditis)
- Gastrointestinal disease
- Interstitial renal and lung diseases
- Involvement of the biliary tract and pancreas



Blood vessels become constricted with cold temperatures or stress leading to blue discoloration. Once the blood vessels undergo vasodilation, the skin becomes red.

Sjogren's Syndrome

- **Diagnosis Test**

- Quantitative salivary gland scintigraphy
- Whole sialometry
- Schirmer test
- Ocular staining
- Blood test
- **Labial salivary gland biopsy**

- **Focus score >1 per 4mm²**

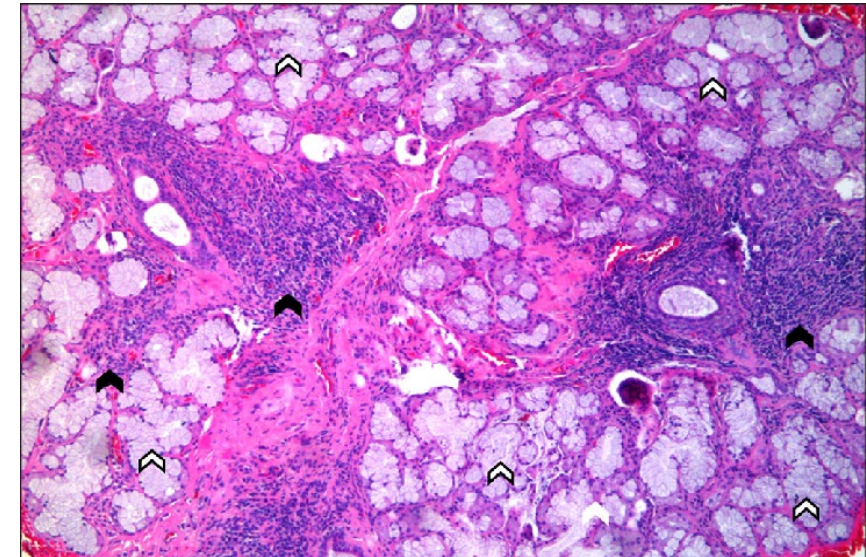
Diagnostic Laboratory Values

- Anti-SSA (anti-Ro), Anti-SSB (anti-La), Rheumatoid Factor (RF)
- Decreased WBC and increased globulins

Labial salivary gland biopsy
Clinical photo with histology



J Oral Pathol Med. 2021 May;50(5):485-491.



Sjogren's Syndrome-Dental Considerations

- Increase caries risk
 - Topical fluoride
- Oral candidiasis
 - Antifungal medication
- Burning sensation of the oral mucosa
- Poor fit of dental prosthesis

Sjogren's Syndrome

- **Prognosis**
 - Patients experience a lot of difficulty with dry eyes and mouth
 - 40 times more likely to develop lymphoma
 - Mucosal associated lymphoid tissue lymphoma (MALT Lymphoma)
- **Treatment**
 - Artificial tears
 - Sialagogues
 - Biotene and Salivasure
 - Cevimeline and Pilocarpine (prescribed cholinergic agonist medications)
 - Glucocorticoids
 - Disease-modifying antirheumatic drugs (DMARDs)
 - Biologics



Lupus

- Most common vascular or connective tissue disease in U.S.
- Immune-complex deposition in small blood vessels
- **Systemic Lupus Erythematosus (SLE)**
 - Most severe complications
- **Discoid Lupus Erythematosus (DLE)**
 - Predominantly affects the skin
- Medication-Induced Lupus
- Neonatal Lupus

Lupus

- **Systemic Lupus Erythematosus**

- Female predilection

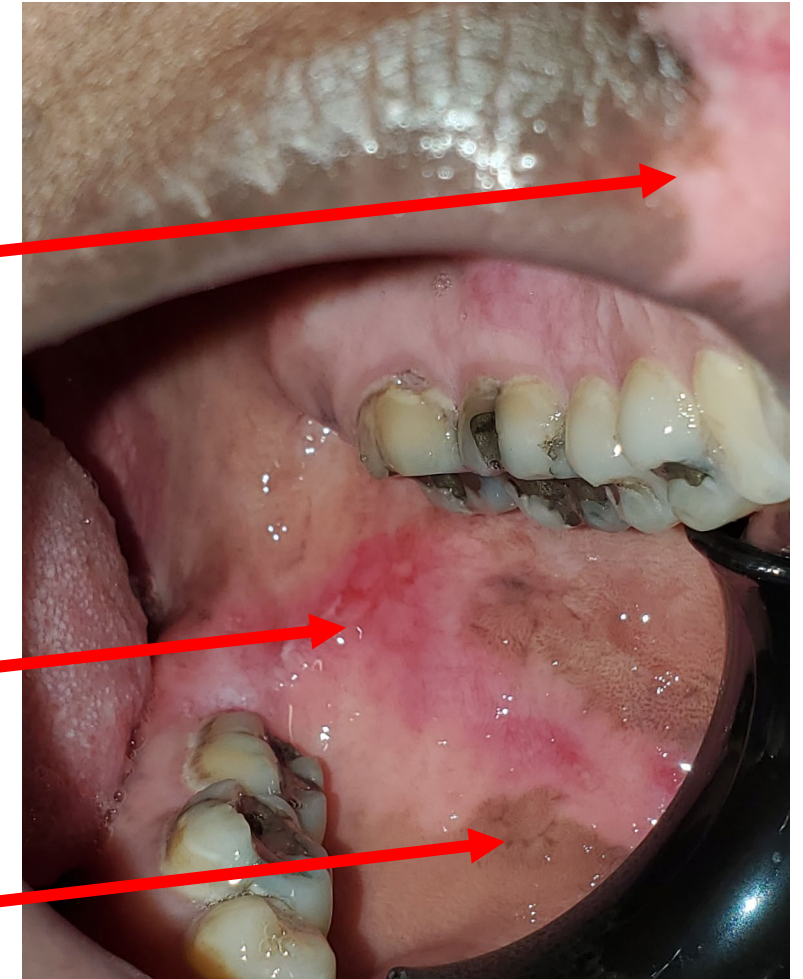
- **Clinical presentation**

- Arthritis
 - Fever
 - Malar rash
 - Photosensitivity
 - Oral erosive lesions
 - Renal disease
 - Cardiovascular disease
 - Pulmonary disease
 - Hematologic disease

Hypopigmentation
of the skin of the
vermillion border
of the lip.

Oral erosive lesion
presenting as an
erythematous area
on the left buccal
mucosa.

Medication-induced
hyperpigmentation of
the oral mucosa



Lupus

- **Discoid Lupus Erythematosus**
 - Circular sores and scars on the skin of the ears, nose, and cheek bones
 - **Malar rash – “Butterfly rash”**



SEMINAR | VOLUME 384, ISSUE 9957, P1878-1888, NOVEMBER 22, 2014



Courtesy of Dr. Edmund Cataldo

Lupus-Dental Considerations

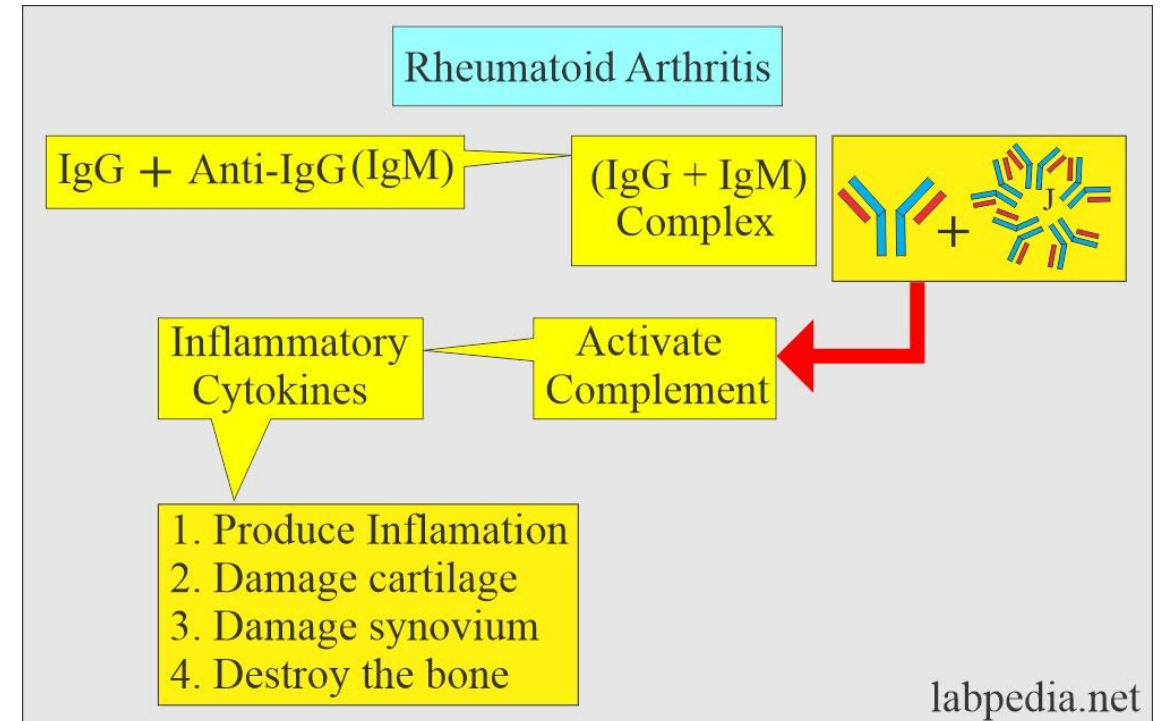
- Pre-treatment medical consultation
- Xerostomia
- Painful ulcers of the mouth
- Risk of infection due to immunosuppression
- Abnormal bleeding due to thrombocytopenia
- Drug-induced pigmentation of the oral cavity
- Prophylactic antibiotics
 - In patients with valvular disease of the heart

Diagnosis

- Clinical symptoms
- **Serologic Test**
 - Immunofluorescent studies
 - Antinuclear antibodies
 - Anti-double stranded DNA
 - Antiphospholipid antibodies
 - **Anti-Sm antibodies**

Rheumatoid Arthritis

- Peak incidence at ages 35-50 years of age
- Female predilection
- Genetic susceptibility with an external trigger
 - Tissue marker: HLA-DRB1
- **Pannus** - Hypertrophy of the synovial lining of the joints
- Inflammatory destruction of synovial lining, cartilage, and bone
 - Formation of IgG antibodies (IgM-rheumatoid factor) leading production of IgG & IgM immune complexes
 - The complexes activate the immune system



Clinical Findings

- Formation of subcutaneous nodules in the joints of the hands, feet, knees, elbow, and ankles
 - Rheumatoid nodules
- Painful and stiff joints
- Fatigue and weakness
- Radiographic bony changes



www.pcds.org.uk/clinical-guidance/rheumatoid-nodules

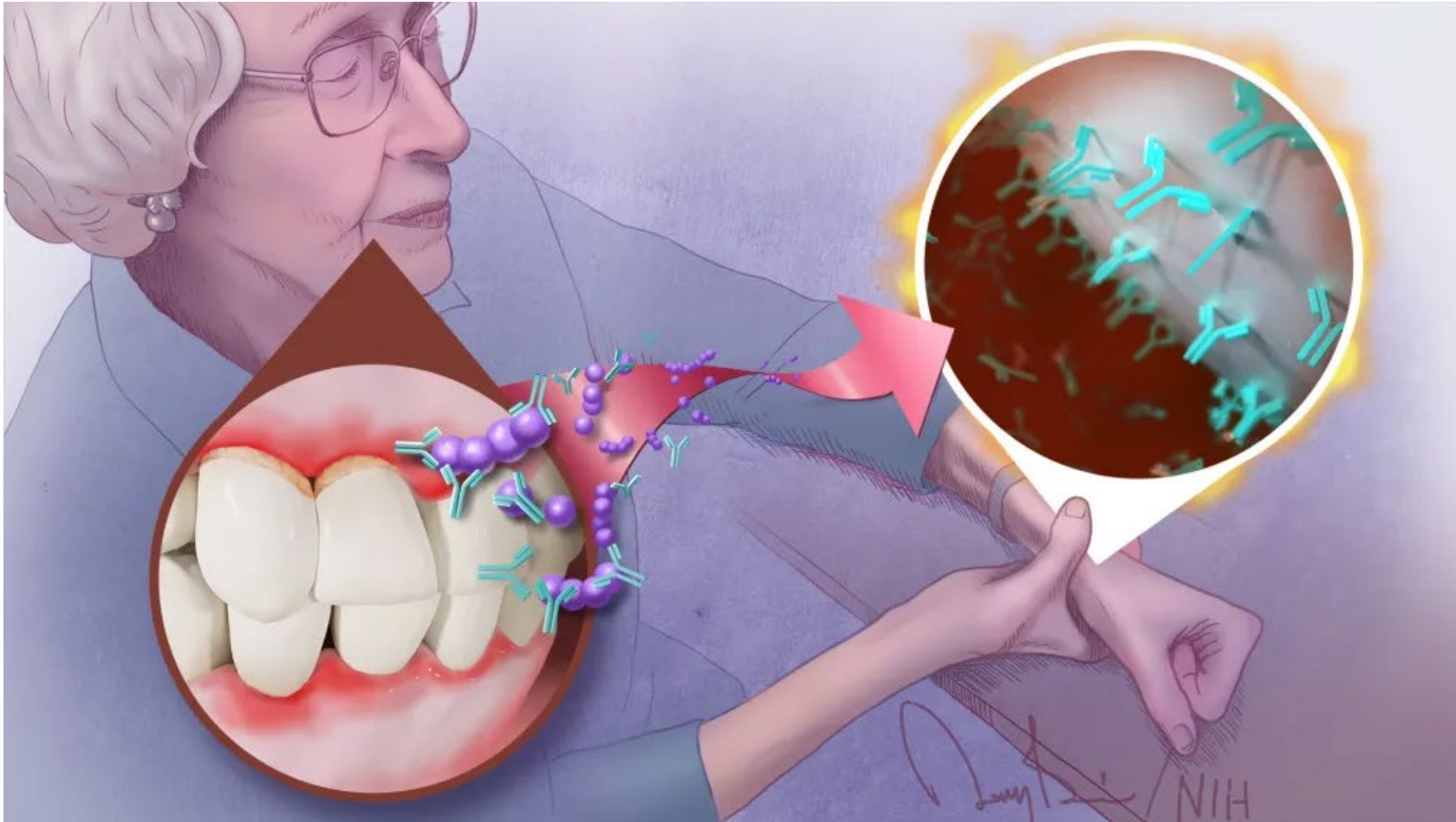
Rheumatoid Arthritis

- Criteria for Diagnosis
 - **Evaluation (4 or more of the following)**
 - Morning joint stiffness lasting at least 1 hour
 - Arthritis of three or more joint areas
 - Arthritis of the hand joints
 - Symmetric arthritis
 - Rheumatoid nodules
 - Abnormal amounts of serum rheumatoid factor
 - Radiographic changes

Rheumatoid Arthritis affecting the TMJ



Connection between Dental Health and RA



Rheumatoid Arthritis

- **Dental Considerations**

- Chair positioning
- Adverse reactions associated with medications
 - Dry mouth, bleeding, **oral ulcers (Methotrexate-induced)**
- Avoid elective dental treatment for at least 30 days after joint replacement
- Helping patients with poor manual dexterity
- Temporomandibular joint disorder

- **Treatment**

- NSAIDs, corticosteroids, DMARDs, and biologics, joint replacement

Antibiotic Coverage and Joint Replacement

Table 20.5 Textbook

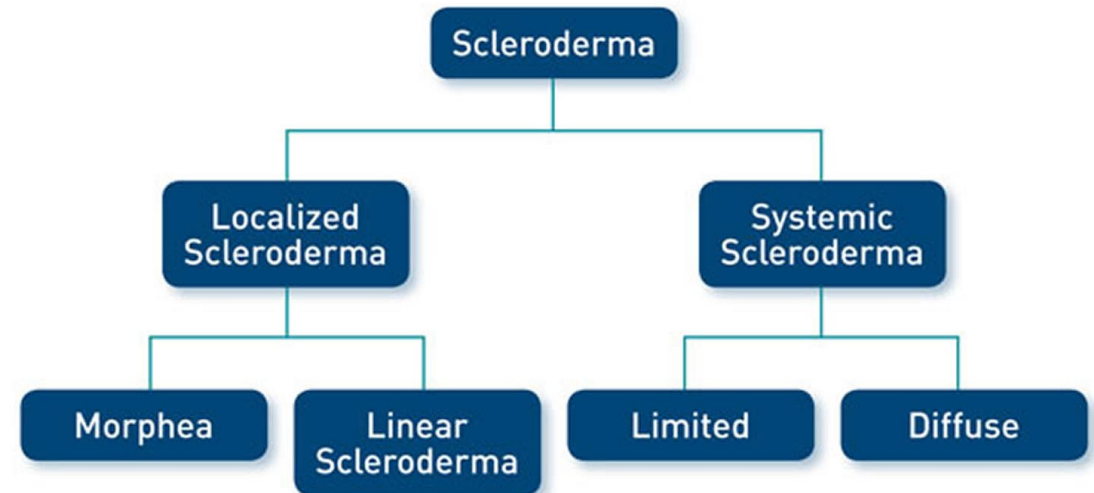
- Antibiotic prophylactic coverage not recommended prior to dental procedures in those with joint replacements.
 - The need for antibiotic coverage should be discussed with the orthopedic surgeon
 - The individual patient's health and preferences should be taken into consideration prior to recommending antibiotic prophylaxis
 - Immunocompromised patients

Progressive Sclerosis (Scleroderma)

- Thickening of the skin associated with tissue fibrosis and inflammation
- Affects visceral organs, heart, lungs, and kidneys
- Vascular dysfunction
- Female predilection
- 30-50 years the range of onset

Progressive Sclerosis (Scleroderma)

- **Localized Scleroderma**
 - Affects the skin of the hands and face
- **Systemic Scleroderma**
 - Affects large areas of the skin and organs
 1. Limited (CREST Syndrome)
 2. Diffuse Disease



Limited Disease AKA CREST Syndrome

The limited symptoms of scleroderma are referred to as **CREST**

Calcinosis- calcium deposits in the skin



Raynaud's phenomenon- spasm of blood vessels in response to cold or stress



Esophageal dysfunction- acid reflux and decrease in motility of esophagus



Sclerodactyly- thickening and tightening of the skin on the fingers and hands



Telangiectasias- dilation of capillaries causing red marks on surface of skin



Sclerodactyly and Calcinosis Cutis



www.aocd.org/page/SystemicSclerosis



Calcinosis. Credit: Mohammad2018, CC BY-SA 4.0, via Wikimedia Commons

Raynaud's Phenomenon



Hughes M et al. Best Prac Res Clin Rheum. (2016)

Telangiectasias of the face, finger, and labial mucosa



Diffuse Disease

- **Bone and muscle changes**
 - Pain, stiffness, and swelling of the fingers and joints
 - Acro-osteolysis –erosion of the distal portion of fingers and toes
- **Pulmonary symptoms**
 - Dry cough and shortness of breath
- **GI symptoms**
 - Reflux, dysphagia, constipation/diarrhea

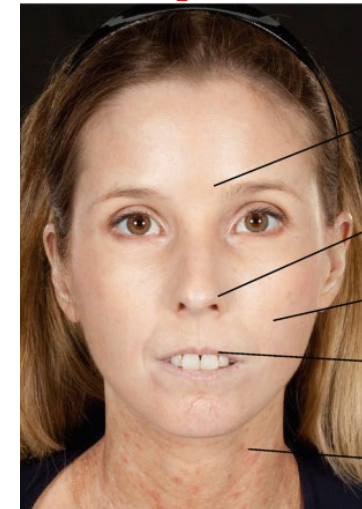
Scleroderma

- Facial changes

- Tightness around the mouth (mask-like and mouse facies)
- Trismus
- “Pursed lips”



FIGURE 4 Scleroderma: characteristic facies.
Source: American College of Rheumatology.



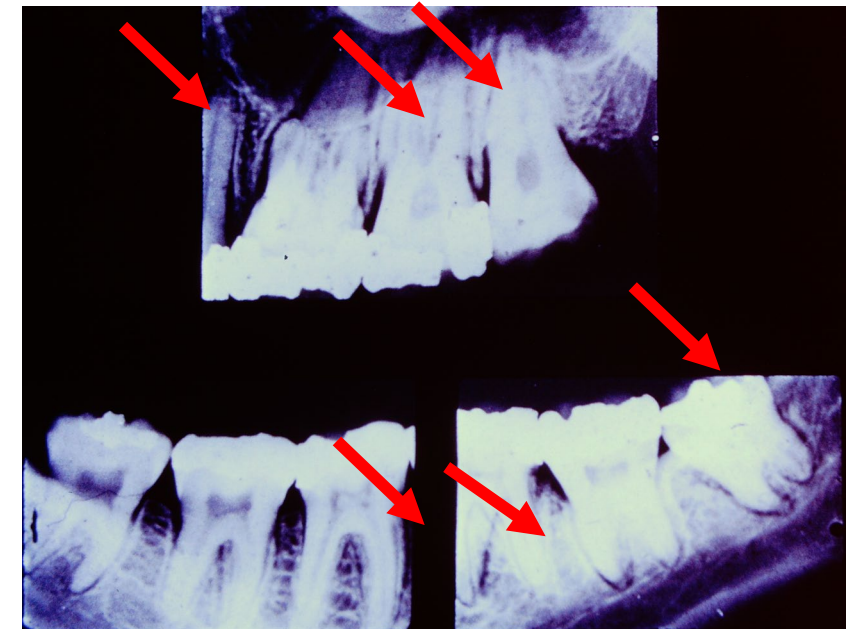
Almadori, A., Butler, P.E.M. (2022). Treatment of Scleroderma with Fat Grafting, PRP, and Adipose-Derived Stem Cells. In: Kalaaji, A. (eds) Plastic and Aesthetic Regenerative Surgery and Fat Grafting. Springer, Cham.

Scleroderma

- Radiographic Changes
 - Collagen deposition can cause retraction of the jawbone
 - Diffuse widening of the PDL



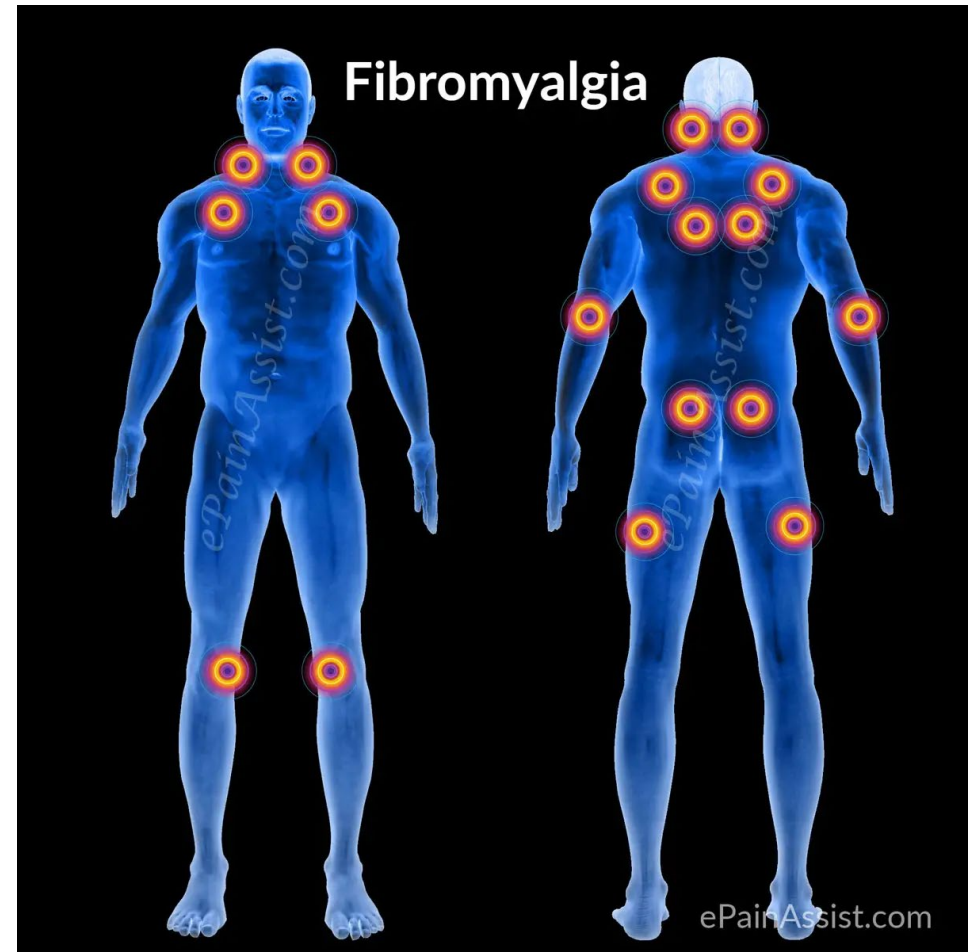
Progressive Systemic Sclerosis - an extensive manifestation of Scleroderma. *South African Dental Journal* , 70(4), 156-158. Retrieved May 09, 2023



Courtesy of Dr. Edmund Cataldo

Fibromyalgia

- Widespread musculoskeletal pain
- Depression, anxiety, cognitive, and sleep disturbances
- Female predilection
- Considered to be one of the chronic pain disorders



Fibromyalgia

- **Criteria for diagnosis**
 - Widespread pain persisting for > 3 months
 - Stiffness
 - Fatigue
 - Cognitive difficulties
 - Other unexplained symptoms

Fibromyalgia

- **Oral manifestations**
 - Temporomandibular joint disorder
 - Xerostomia
 - Caries
 - Candidiasis

Resources

